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Project: Beginner's Toolkit for C# in Game Development

Scripting My First Game with C# and Godot

Technology: C# (Programming Language) via Godot Engine 4.x (.NET Edition)

1. Title & Objective

a) What technology did you choose?

C# and the Godot Game Engine.

b) Why did you choose it?

C# is a professional, high-performance language and Godot is open-source, lightweight, and has a very modern .NET implementation.

c) What's the end goal (e.g., render a styled UI component)?

To successfully link a coding environment (VS Code) to a game engine (Godot) and create a "Player Controller" script that moves a 3D or 2D object.

2. Quick Summary of C# and Godot

a) What is it?

C# acts as the "brain" of the game and provides the instructions while Godot acts as the body and provides the visual world.

b) Where is it used?

In Godot, every object is a "Node."

In C# we attach scripts to Nodes to tell them how to behave when a player presses a button.

c) One real-world example?

Games like Cassette Beasts show that Godot and C# can create high-quality games.

3. System Requirements

a) OS: Windows 10/11 (or macOS/Linux).

b) Hardware: 4GB+ RAM (8GB recommended).

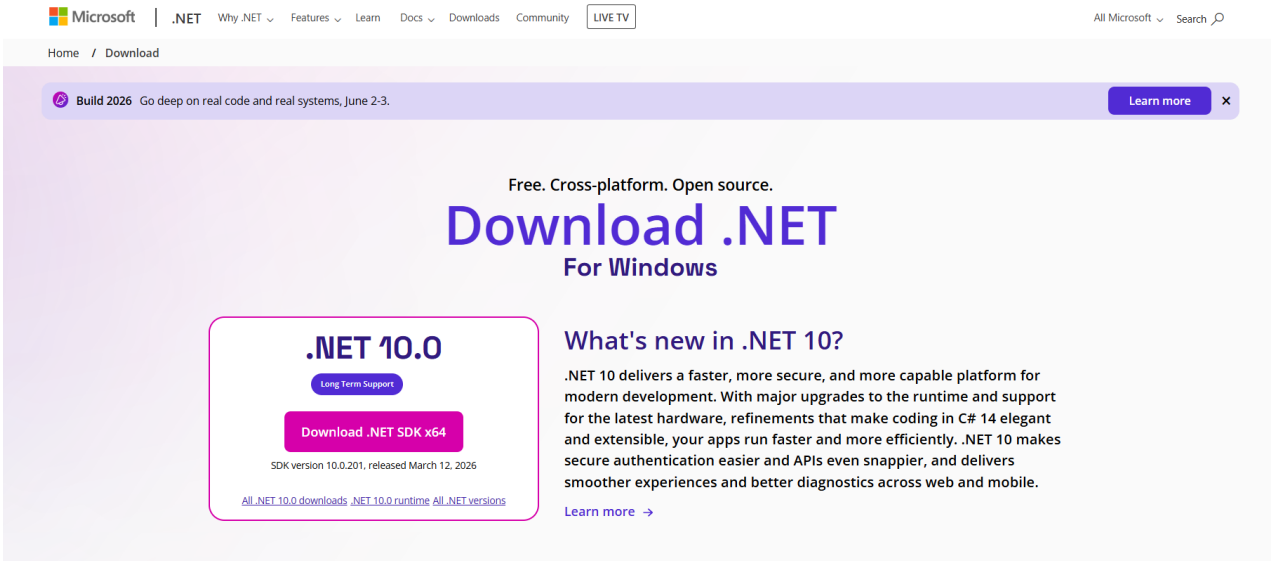
c) SDK: .NET 10.0 SDK (Installed to compile C# code).

d) Editor: Godot Engine 4.x - .NET Edition (Note: The Standard version will not work for C#).

e) IDE: Visual Studio Code (VS Code).

4. Installation & Setup Instructions

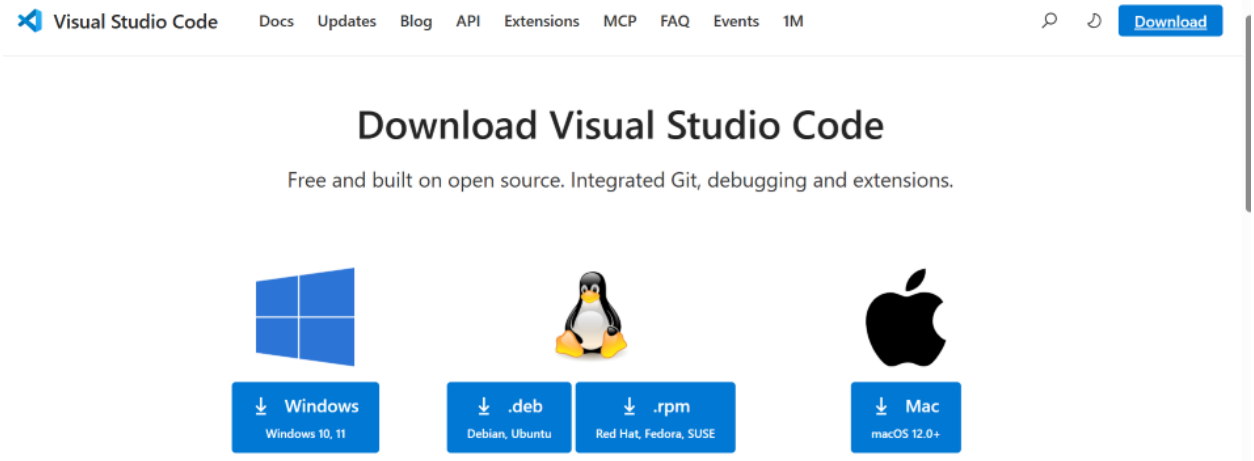
1. **Install the SDK:** Download and install **.NET 10.0** from Microsoft.



2. **Download the Engine:** I learned that there are two versions of Godot. I had to specifically download the **.NET Edition** to use C#.



3. **Download the IDE VS Code**: Download and install the VS Code



4. **Find the Editor Menus:** I initially looked for settings in the "Project Manager", but I learned you must Create/Open a Project first to see the main Editor menus.

5. **Link VS Code:** In the Godot Editor, I went to Editor then Editor Settings. Navigated to Dotnet then Editor and selected Visual Studio Code as the External Editor.

6. **Create the Solution:** To help VS Code understand Godot, I went to Project then Tools then C# then selected Create C# Solution.

5. Building the Game: Step-by-Step Example

Step 1: Create your "Player" Object

In Godot, every object is a Node.

1. In the Scene tab (top left), click the + button.
2. Search for Sprite2D and click Create.
3. Right-click the new Sprite2D node and select Rename. Name it Player.

Step 2: Add an Image (So you can see it move)

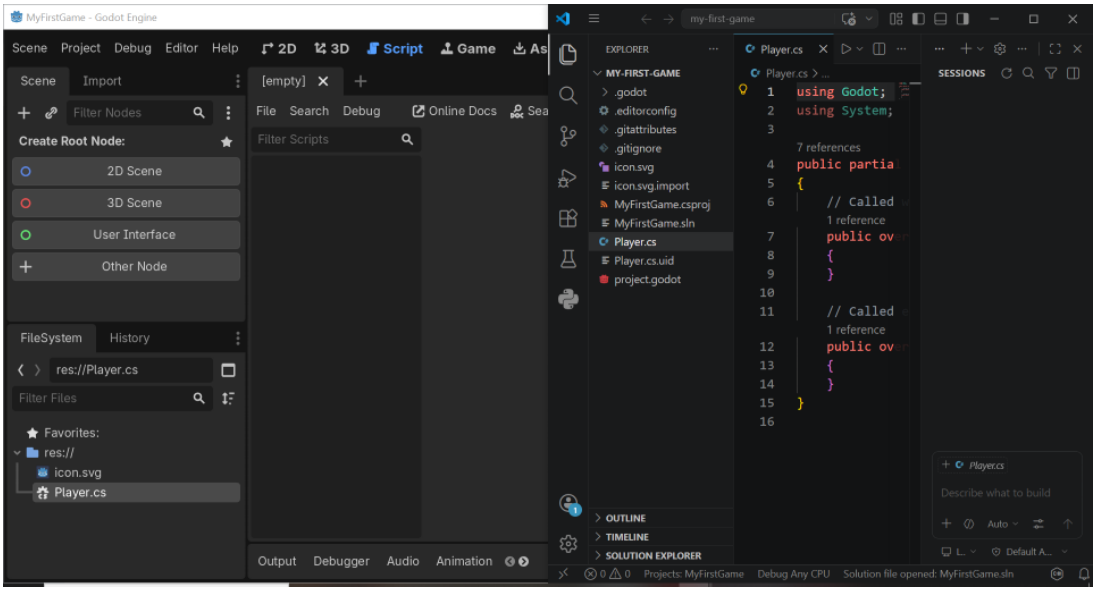
1. Look at the FileSystem tab (bottom left). You should see a file called icon.svg
2. Click and drag icon.svg over to the Inspector tab (on the right) and drop it into the Texture slot.Now you should see the Godot logo in the middle of your game window!

Step 3: Attach the C# Script

1. Right-click your Player node in the Scene tab.
2. Select Attach Script.
3. In the popup window:

    - Change Language to C#.
    - Click the folder icon next to Path and name the file Player.cs.

4. Click Create. VS Code should automatically pop open.



Step 4: Write the Code in VS Code

```
using Godot;
using System;

public partial class Player : Sprite2D
{
    [Export]
    public float Speed = 400f;

    public override void _Ready()
    {
        GD.Print("Setup Successful! VS Code and Godot are linked.");
    }

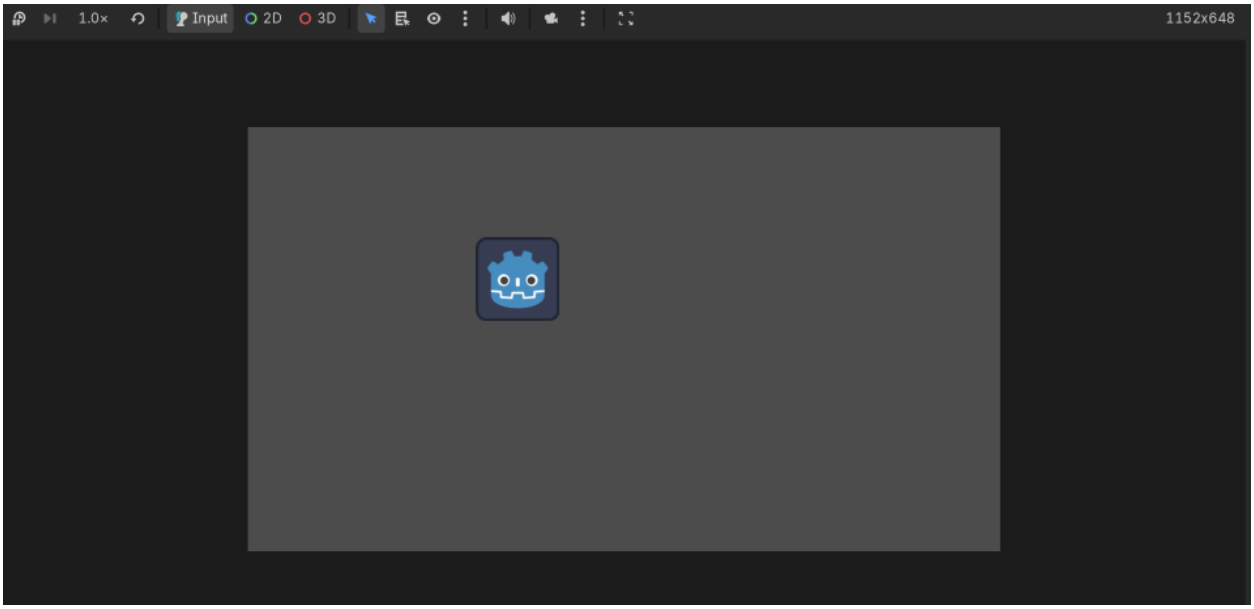
    public override void _Process(double delta)
    {
        Vector2 velocity = Vector2.Zero;

        if (Input.IsActionPressed("ui_right")) velocity.X += 1;
        if (Input.IsActionPressed("ui_left"))  velocity.X -= 1;
        if (Input.IsActionPressed("ui_down"))  velocity.Y += 1;
        if (Input.IsActionPressed("ui_up"))    velocity.Y -= 1;

        // Move the icon based on input and time (delta)
        Position += velocity.Normalized() * Speed * (float)delta;
    }
}
```

Step 5: Build and Play

1. Go back to Godot.
2. Look at the top-right corner of the screen. Click the Build button (it looks like a small hammer or a brick). Wait for it to finish. This turns your C# text into instructions the computer understands.
3. Press the Play button (the triangle icon) or press F5.
4. If it asks you to "Select a Main Scene," click Select Current.
5. Test it: Use your arrow keys. Your Godot icon should now be moving around the screen.



6. AI Prompt Journal

a) Prompt used: Why can't I find the Dotnet menu in settings?
AI's response summary: Gemini explained that I likely had the Standard version of Godot which I fixed by downloading the .NET Edition.
b) Prompt used: Why is my movement jittery?
AI's response summary: Gemini taught me about delta, which ensures movement is smooth regardless of the computer's frame rate.
Your evaluation of its helpfulness?
Using Gemini turned a frustrating setup process into a learning experience. Instead of being stuck for days on a missing menu, I got an immediate explanation of why it was missing which was because I downloaded the wrong version of Godot.

7. Common Issues & Fixes

Issue	Fix
MSBuild not found	Install the .NET SDK and ensure you are using Godot .NET Edition.
No Autocomplete in VS Code	Click Project > Tools > C# > Create C# Solution to sync files.

Issue	Fix
Settings menu is missing	Open a project first; you can't edit these in the Project Manager.

8. References

Godot Official C# Documentation
Microsoft .NET Download Page
Google (2026) *Gemini*. Available at: https://gemini.google.com/